CACTIxT USER GUIDE

1. Package contents

The supplied archive contains the directory **CACTIxT_CodeBL**. In the current package, the CACTIxT C/C++ source files (.cc and .h) are stored directly in this directory together with the Code::Blocks project file and the configuration resources.

- CACTIxT_CodeBL.cbp: Code::Blocks console-application project. The project is configured to use GCC and provides Debug and Release build targets.

- **xmls**: directory containing the XML files used to configure the memory architecture and the target technology. In particular, cache_config_cmos.xml configures CMOS-based analyses. The subdirectories devices, sram_cells, and dram_cells contain technology/device descriptions and memory-cell definitions. The supplied CMOS SRAM-cell files include 6T, 8T, and 10T topologies.

- The source-code entry point is main.cc.

2. Running CACTIxT in Code::Blocks

Open the project file CACTIxT_CodeBL.cbp in Code::Blocks. Before running the program, select the desired build target and verify the program arguments from **Project > Set program's arguments**. For CMOS simulations, the Program Arguments field must contain:

```
-infile ./xmls/cache_config_cmos.xml
```

In the supplied project file, this argument is already associated with the **Debug** target. If the **Release** target is used, set the same argument for that target before execution. Run CACTIxT from the project directory so that the relative XML paths can be resolved correctly.

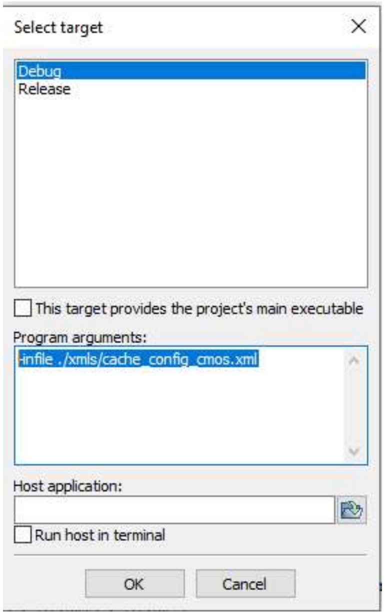

Figure 1. Code::Blocks program arguments for the CMOS configuration.

3. Configuring the memory structure

For CMOS 6T/8T/10T SRAM analyses, edit ./xmls/cache_config_cmos.xml. The line numbers below refer to the files in the supplied archive; if the XML files are subsequently edited, use the indicated XML tags as the authoritative reference.

1) **Select the SRAM-cell topology** at lines 49-51. Exactly one <sram_cell> entry should be active. The supplied file selects cmos_sram_6T.xml by default; uncomment the 8T or 10T entry and comment the others when a different topology is required.

```
<sram_cell>xmls/sram_cells/cmos_sram_6T.xml</sram_cell>
<!-- <sram_cell>xmls/sram_cells/cmos_sram_8T.xml</sram_cell> -->
<!-- <sram_cell>xmls/sram_cells/cmos_sram_10T.xml</sram_cell> -->
```

2) **Select the technology/device XML file** in the <devices> block at lines 37-46. To use another CMOS technology, replace cmos_14nm_std.xml with an XML file that follows the same structure. The selected SRAM-cell XML file also contains <device_type> references for its transistors; when introducing a new technology file, update those references consistently.

3) **Set the technology node** at line 15 using <technology_node>. The value is expressed in micrometers; for example, 0.014 corresponds to a 14 nm technology.

4) **Set the operating-voltage regime** at line 18. The configuration parser accepts the exact strings super-threshold and near-threshold. The corresponding supply values are read from the technology XML file under <vdd><super_threshold> and <vdd><near_threshold>, respectively. In the supplied cmos_14nm_std.xml, these values are 0.8 V and 0.6 V.

5) **Set the operating temperature** at line 21 using <temperature>. The current parser accepts values from 300 K to 400 K in 10 K increments.

6) **Set the memory size** at line 24 using <cache_size>. The value is expressed in bytes; the configuration file specifies a power-of-two integer.

7) **Set the input/output bus width** at line 79 using <bus_width>. The value is expressed in bits. The supplied CMOS configuration uses 256 bits.

4. Technology and variability parameters

The device/technology files are stored in ./xmls/devices/. The supplied cmos_14nm_std.xml provides the reference structure for a planar CMOS technology file.

- **ON-resistance sensitivity:** lines 174-190 contain the <ron_sensitivity> block. Populate the NMOS <S_Rn> and PMOS <S_Rp> entries with the sensitivity values obtained from device-level electrical simulations for the nominal threshold voltage and 1 um-wide devices, for each supply-voltage value of interest. At run time, CACTIxT selects the entry corresponding to the active supply voltage.

- **Threshold-voltage variability:** line 193 contains <sigma_vth>. Set this value to the standard deviation of the threshold voltage, expressed in volts. The supplied 14 nm file uses 0.020 V.

When creating a new technology XML file for the current CACTIxT source tree, preserve the expected XML structure, including the <ron_sensitivity> and <variability> blocks, because these fields are read by the technology parser.

5. Program entry point and output files

Within the Code::Blocks project, the main source file is main.cc. The program accepts the input configuration through the -infile option and passes the selected XML file to the CACTIxT interface.